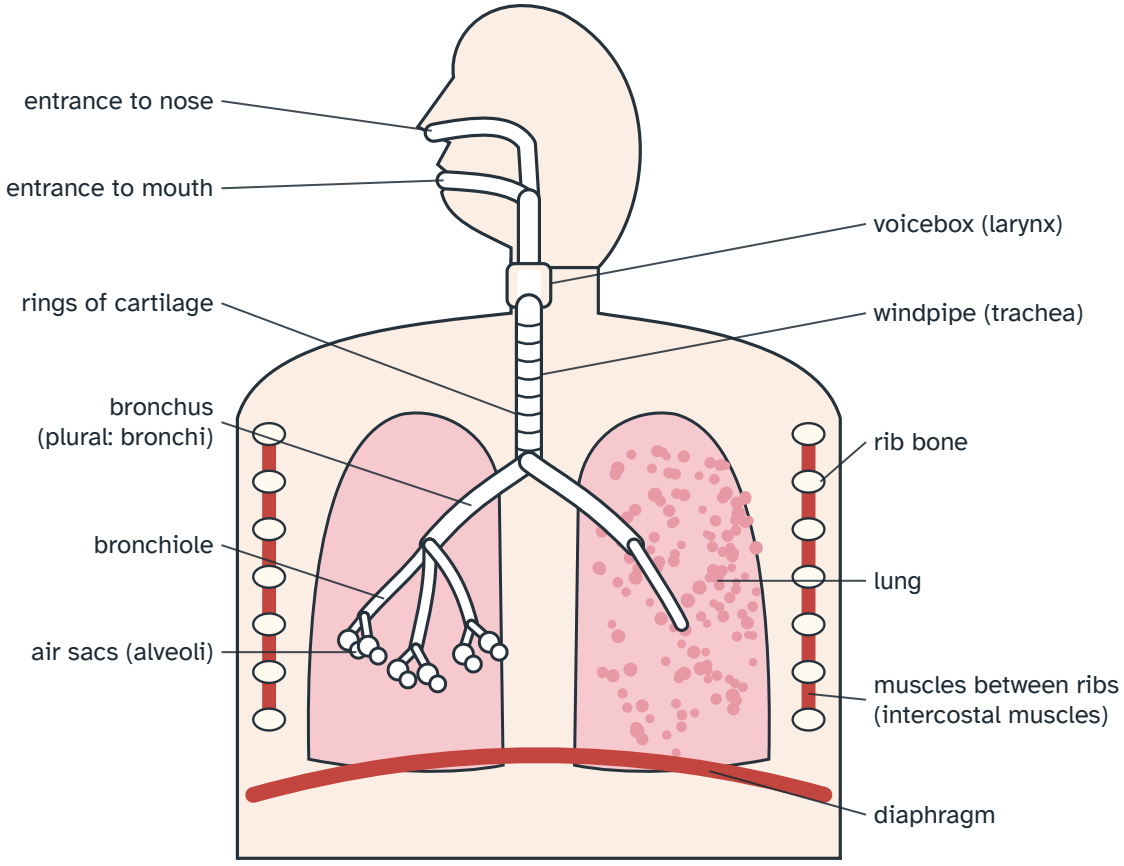
Diagrams to Know

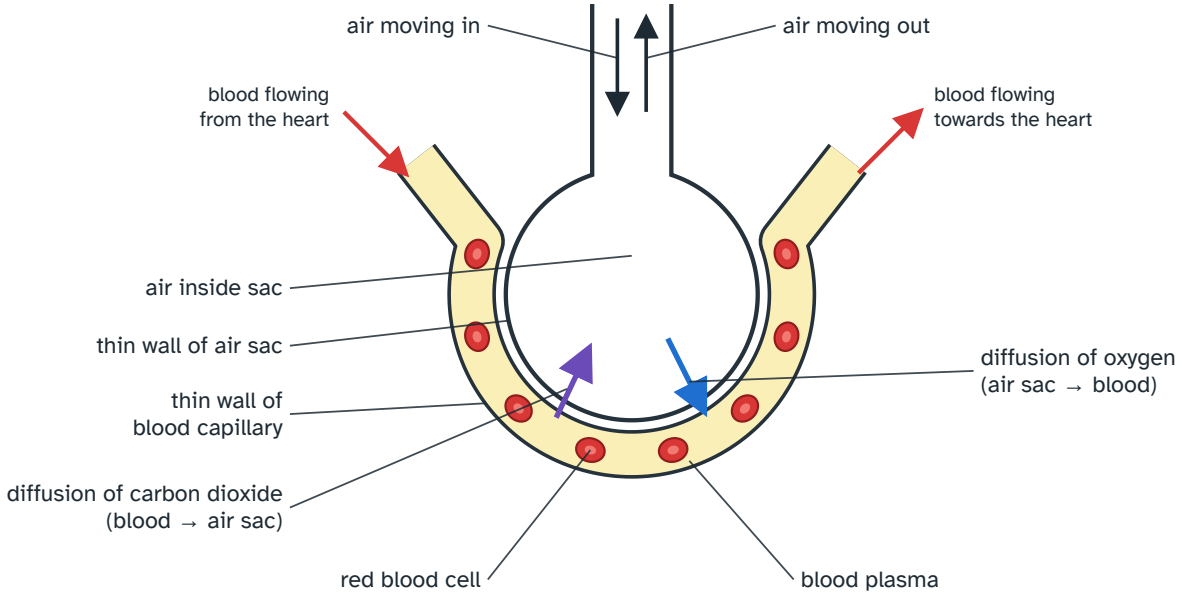
Every labelled diagram from the two chapters, redrawn clearly. Practise drawing and labelling them; test yourself on the last two pages.

1.1 The human respiratory system



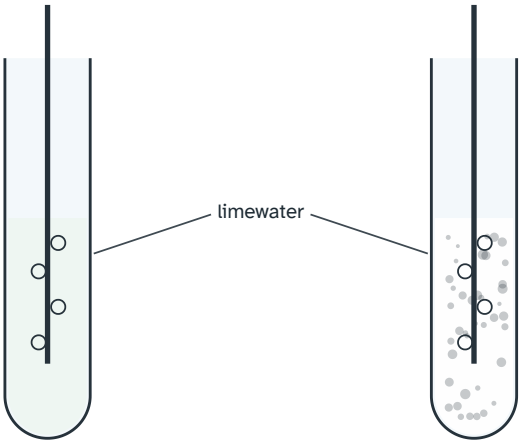
- Air path: nose or mouth → larynx → trachea → bronchus → bronchiole → air sac.
- Rings of cartilage keep the trachea open.
- The intercostal muscles are between the ribs; the diaphragm is under the lungs.

1.2 An air sac and a blood capillary



- Walls of the air sac and the capillary are each ONE layer of very thin cells.
- Oxygen diffuses from the air sac into the blood; carbon dioxide diffuses from the blood into the air sac.

1.2 Limewater test



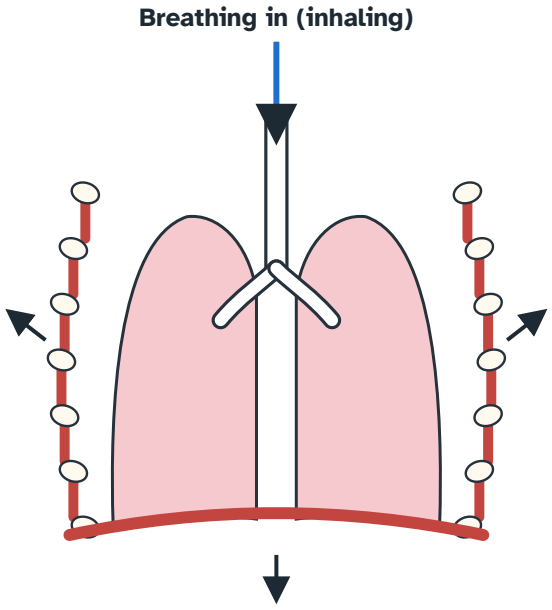
inspired air
limewater stays clear longer

expired air
limewater goes cloudy first

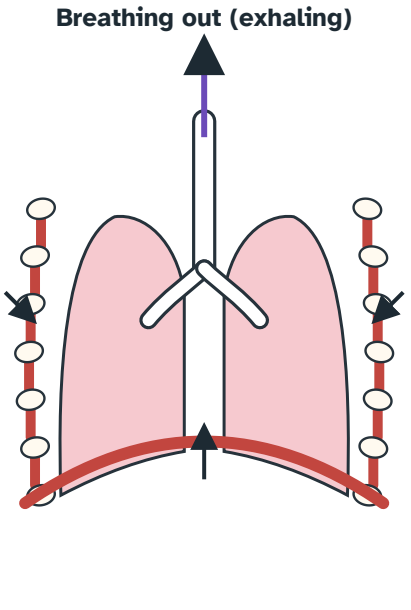
Carbon dioxide turns limewater cloudy, so expired air contains MORE carbon dioxide.

- Carbon dioxide makes limewater cloudy.
- Expired air contains more carbon dioxide than inspired air.

1.3 Breathing in and breathing out



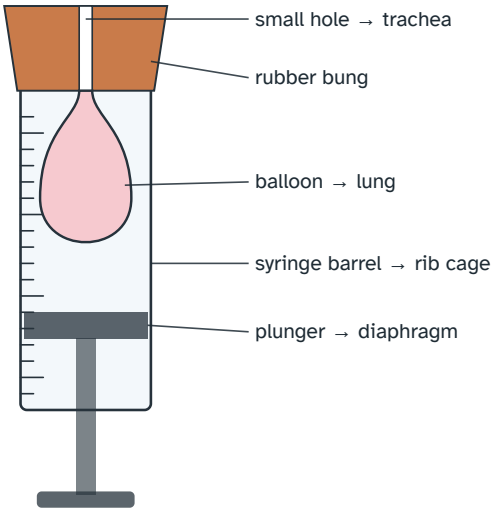
intercostal muscles CONTRACT: ribs move up and out
diaphragm CONTRACTS: moves down (flattens)
chest volume ↑ · pressure ↓ · air moves IN



intercostal muscles RELAX: ribs drop down and in
diaphragm RELAXES: back to its domed shape
chest volume ↓ · pressure ↑ · air squeezed OUT

- In: muscles contract, ribs up and out, diaphragm down, volume up, pressure down, air in.
- Out: muscles relax, ribs down and in, diaphragm domed, volume down, pressure up, air out.

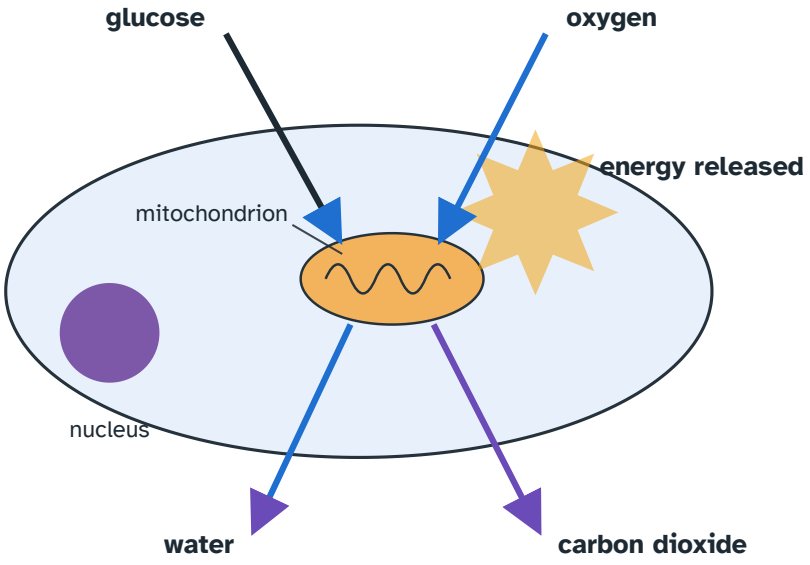
1.3 Syringe model of breathing



Pull the plunger out: volume ↑, pressure ↓, balloon inflates

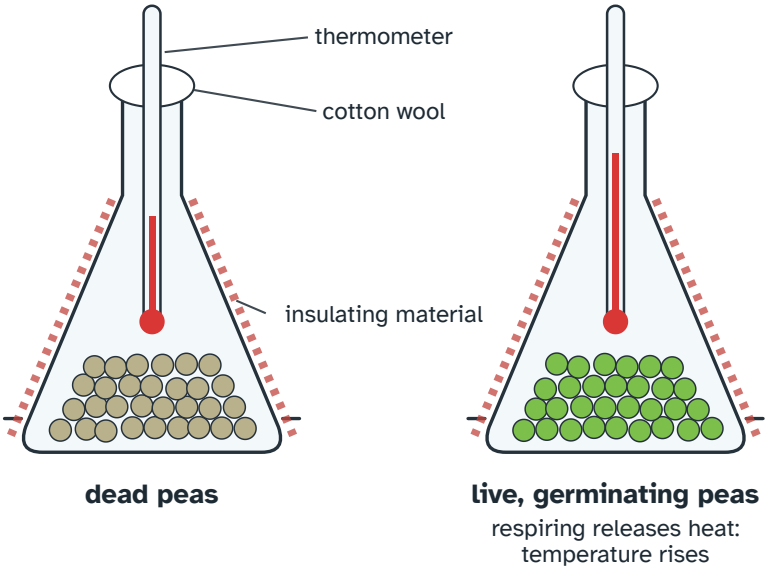
- Balloon = lungs, hole = trachea, plunger = diaphragm, barrel = rib cage.
- Weakness of the model: the barrel (rib cage) does not move.

1.4 Respiration in a mitochondrion



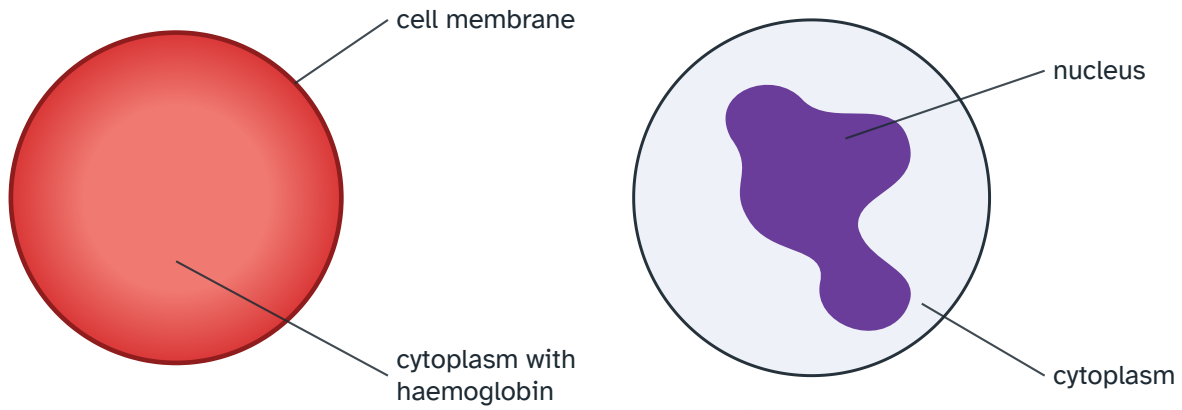
- glucose + oxygen → carbon dioxide + water
- Energy is released a little at a time; some becomes heat.

1.4 Respiration in peas



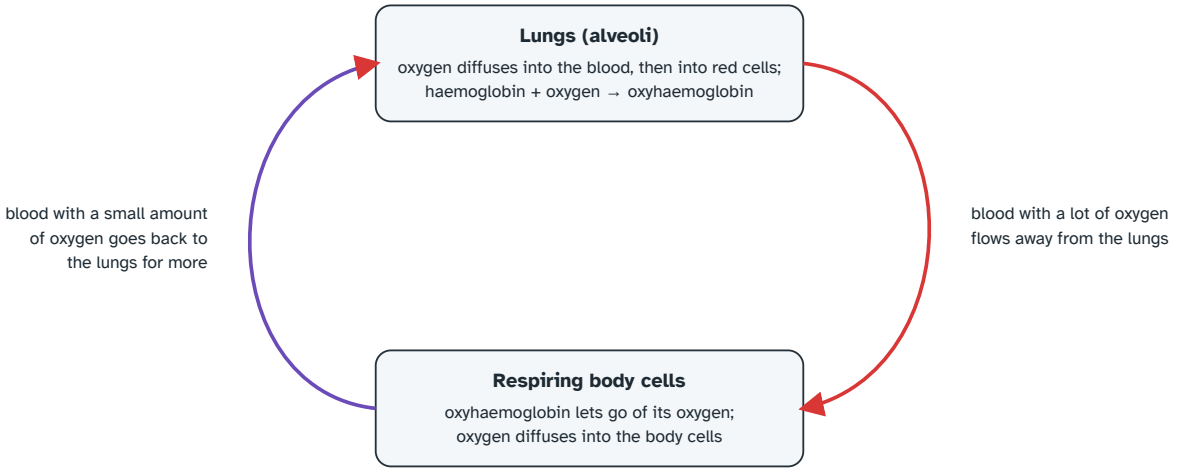
- Changed: dead or live peas. Measured: temperature. Same: everything else.
- Live, germinating peas respire and release heat, so their flask gets warmer.

1.5 A red blood cell and a white blood cell



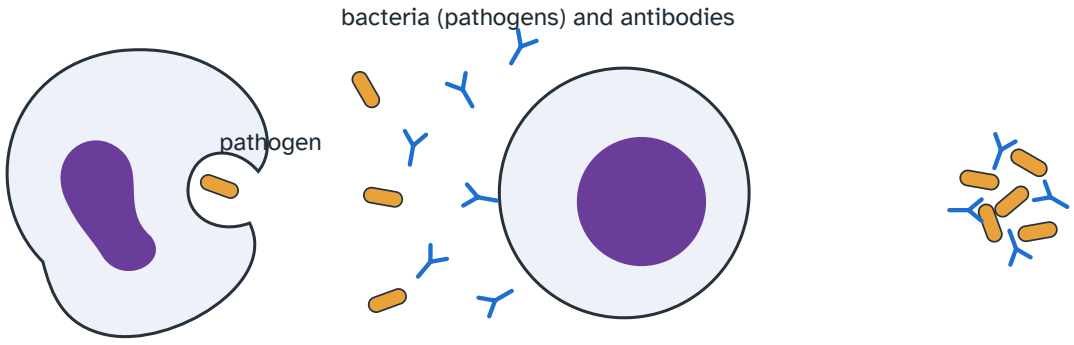
- Red: no nucleus (more room for haemoglobin), no mitochondria (does not use up the oxygen).
- White: always has a nucleus; some kinds are larger than red cells.

1.5 How oxygen is transported around the body



- haemoglobin + oxygen → oxyhaemoglobin (very bright red).
- Oxyhaemoglobin releases its oxygen at respiring cells.

1.5 How white blood cells destroy pathogens



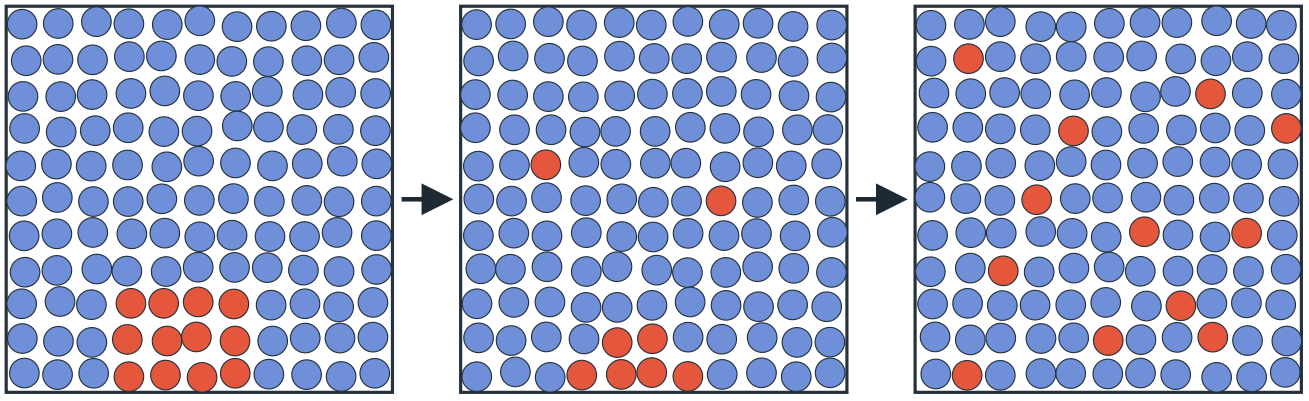
pushes out ‘fingers’ of cytoplasm, captures the pathogen, then makes chemicals that kill and digest it

makes antibodies (Y shapes) that stick to the pathogen and can kill it directly

antibodies glue pathogens together so they cannot move; other white cells kill them

- Pathogens = bacteria and viruses that cause illness.
- A different antibody is needed for each kind of pathogen.

2.1 What happens when sugar dissolves



sugar particles tightly packed in groups (visible)

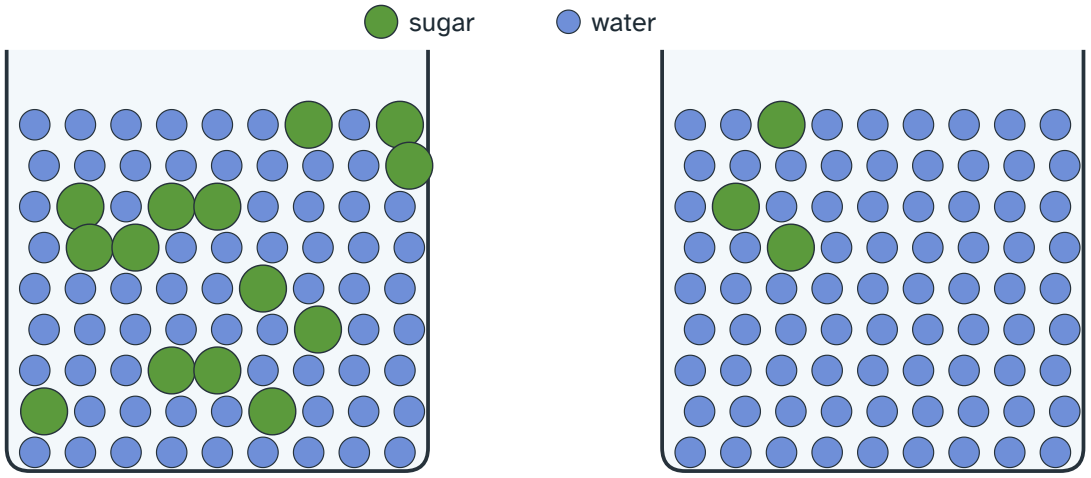
moving water particles bump sugar particles apart

sugar spread out among water: a transparent solution

sugar (solute) water (solvent)

- The sugar is still there: its particles are spread out among the water particles.
- mass of solute + mass of solvent = mass of solution

2.2 Concentrated and dilute solutions

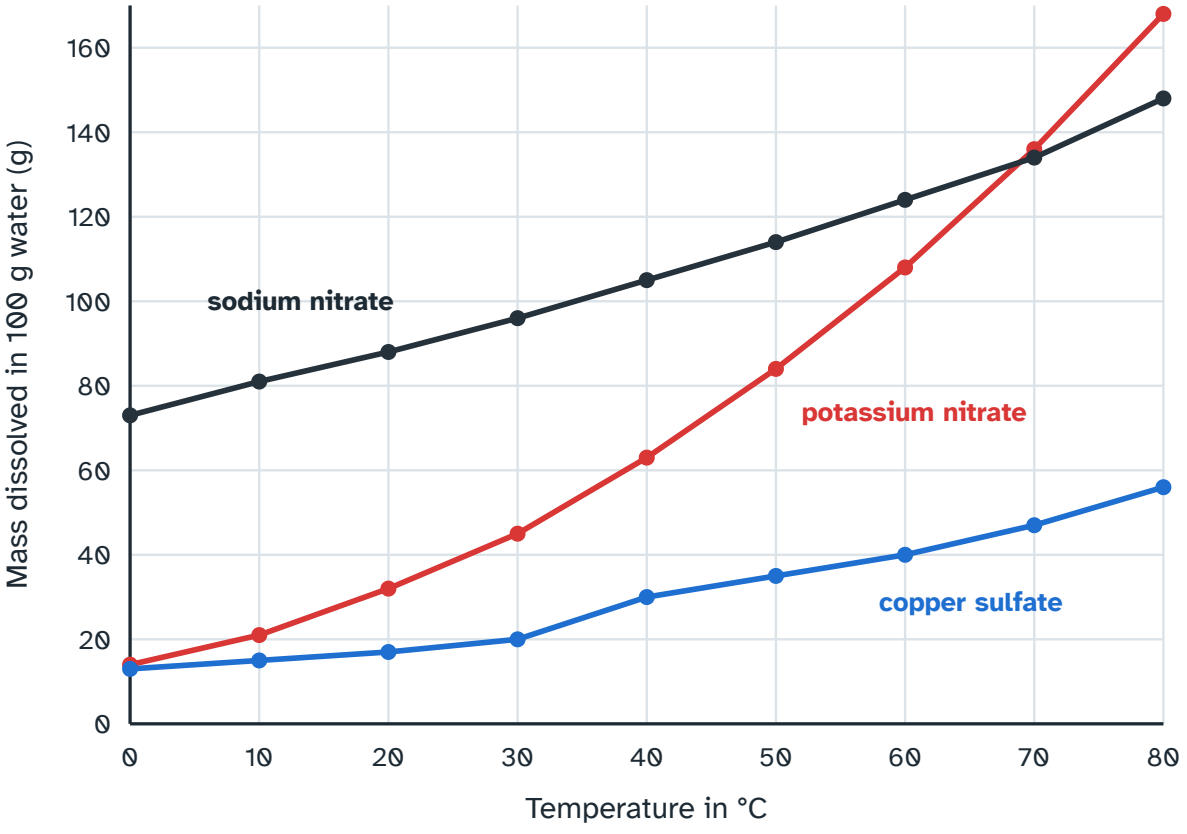


Concentrated
a lot of solute particles

Dilute
fewer solute particles

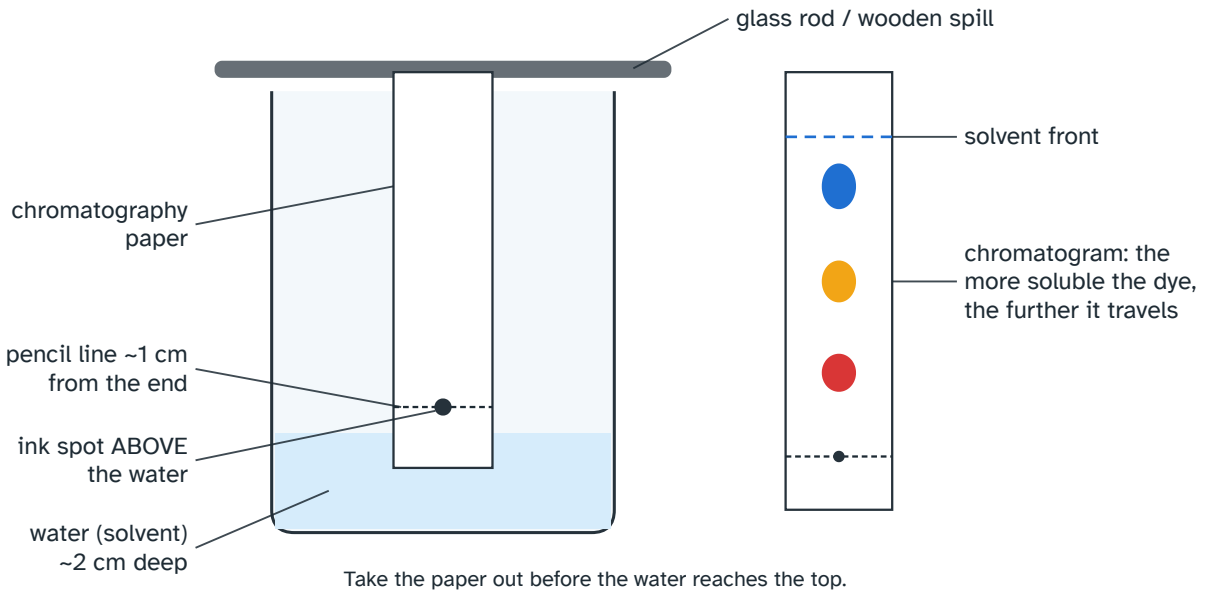
- A concentrated solution has more solute particles dissolved than a dilute one.

2.2 Solubility of three salts at different temperatures



- All three become more soluble as temperature rises.
- The potassium nitrate and sodium nitrate lines cross between 60 °C and 70 °C.

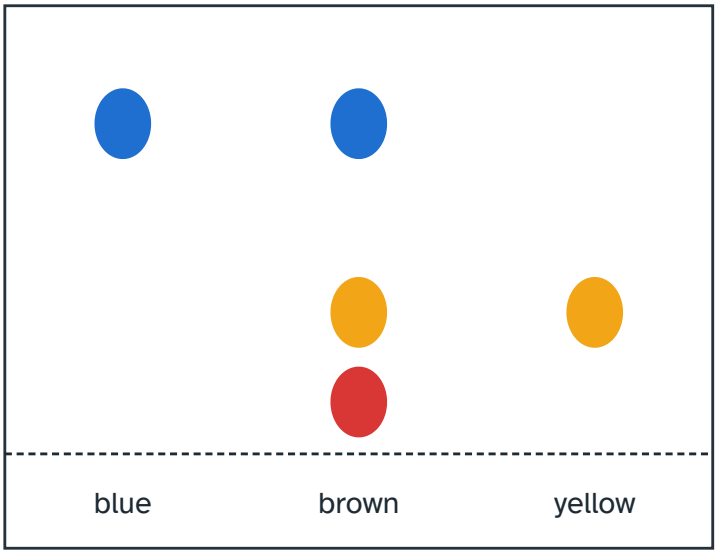
2.4 Paper chromatography



Take the paper out before the water reaches the top.

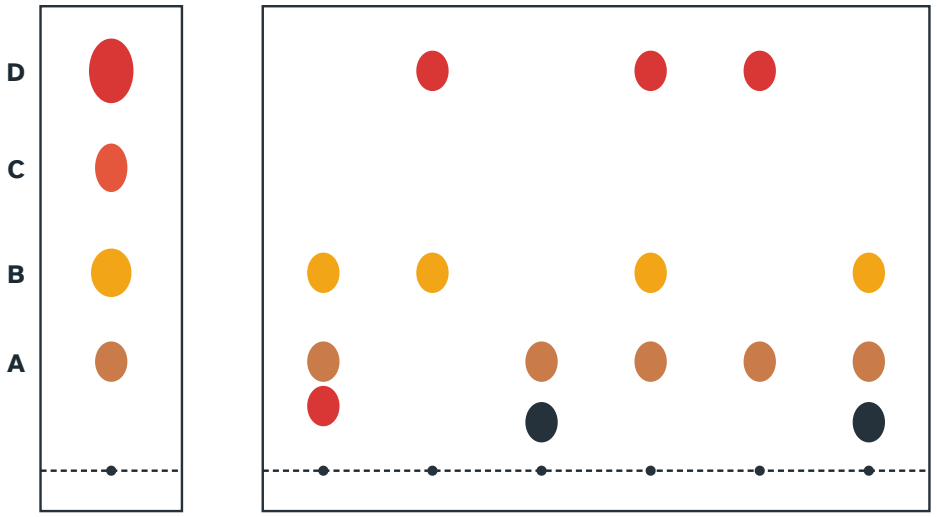
- Pencil line (it does not dissolve), small spot ABOVE the water, take the paper out before the water reaches the top.
- The more soluble the dye, the further it travels.

2.4 Food dye chromatogram



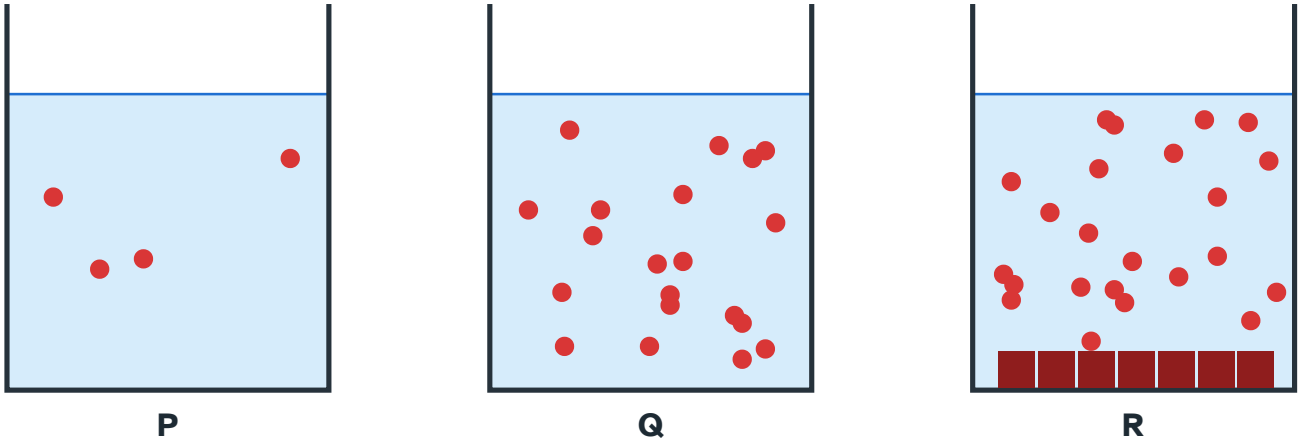
- One spot = pure substance (blue, yellow). Brown is a mixture of blue, yellow and red.
- Blue travelled furthest, so it is the most soluble; red is the least soluble.

2.4 Sunny Red hair dye and permitted dyes



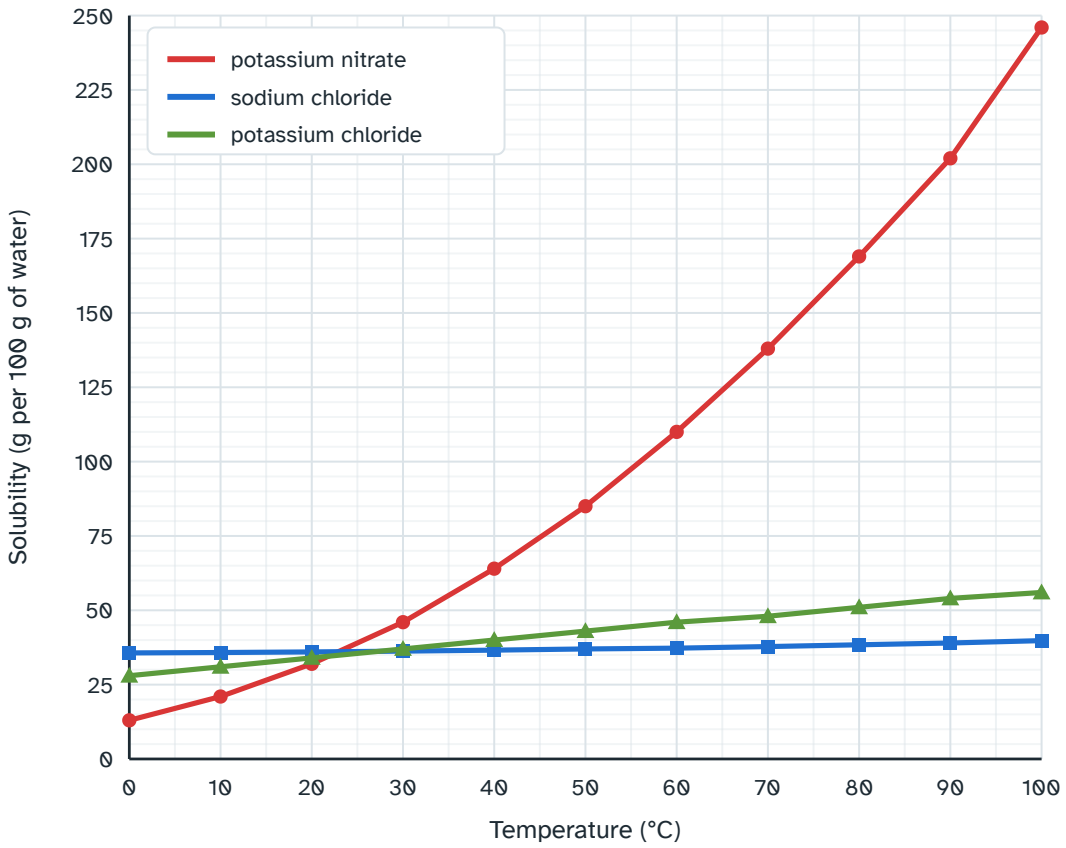
- Spot C does not match any permitted dye, so it is not permitted.
- D travelled furthest, so it is the most soluble.

WS2 Worksheet 2, Figure 1: dilute, concentrated and saturated



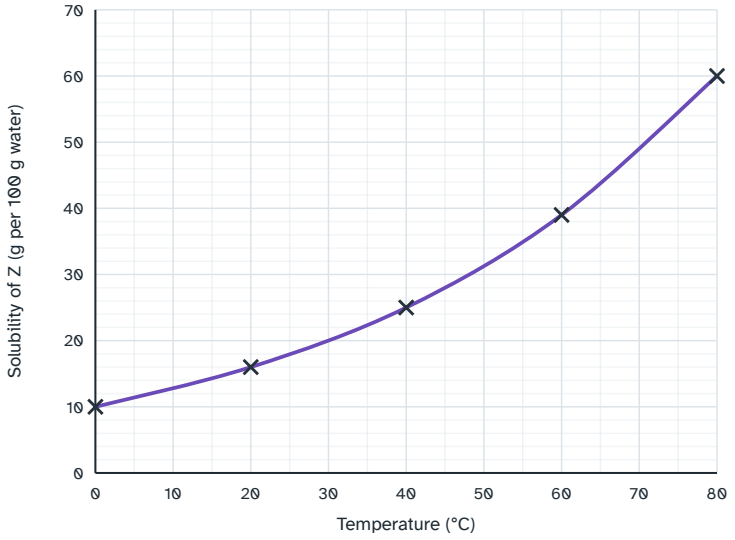
- P: dilute (few solute particles). Q: concentrated but can still dissolve more (unsaturated).
- R: saturated, because undissolved solute is left at the bottom.

WS2 Worksheet 2, Figure 2: solubility curves



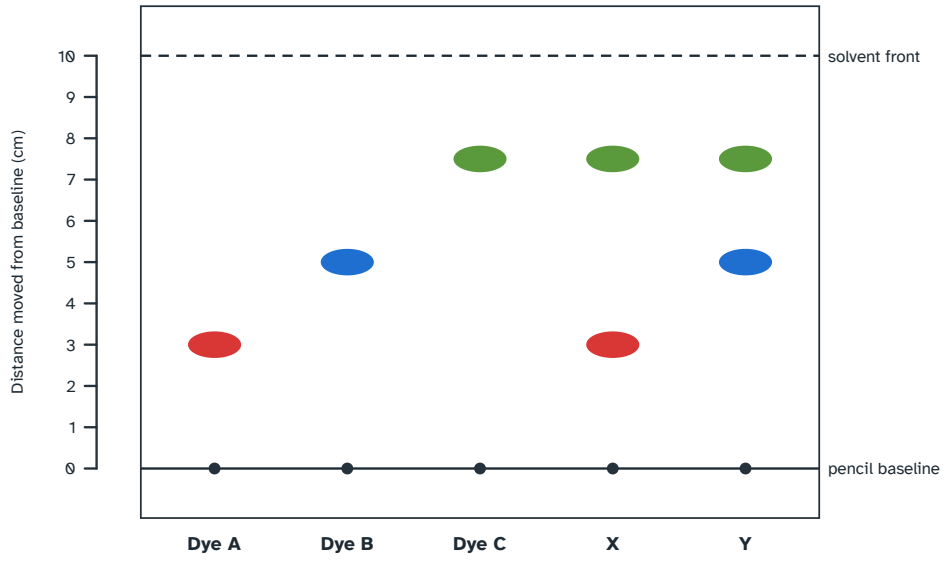
- Potassium nitrate rises steeply; sodium chloride is almost flat.
- Crystals formed on cooling = solubility (hot) − solubility (cold). E.g. 60 °C → 20 °C: 110 − 32 ≈ 78 g.

WS2 Worksheet 2, Q29: solubility of solid Z



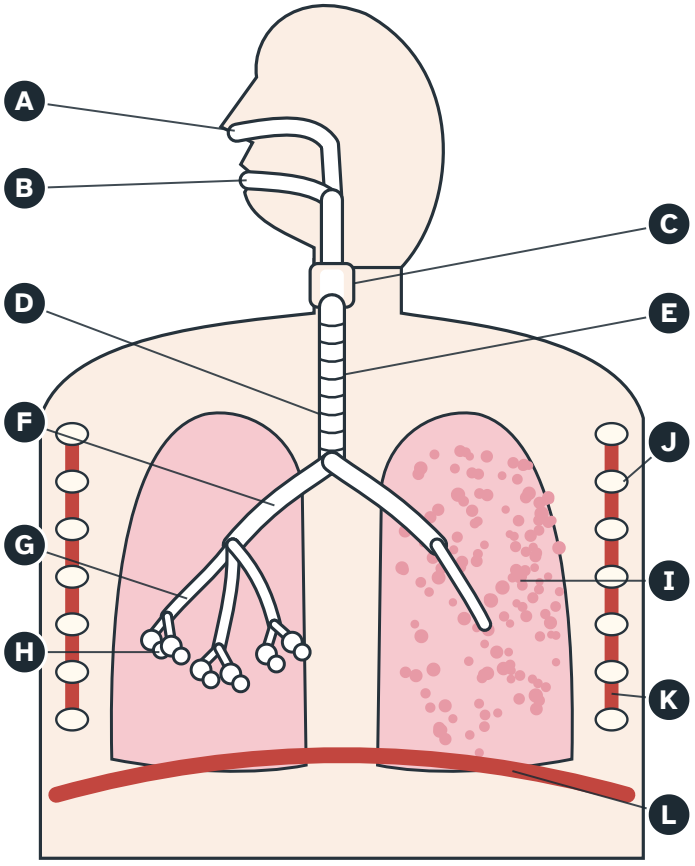
- Plot with neat crosses and draw a smooth curve through the points.
- Z at 50 °C ≈ 31–32 g per 100 g of water.

WS2 Worksheet 2, Figure 4: chromatogram of inks X and Y



- Dyes A, B and C are pure (one spot each). X = A + C. Y = B + C.
- Rf values (front = 10 cm): A = 0.3, B = 0.5, C = 0.75. C is the most soluble.

1.1 Test yourself: label the respiratory system



A _____ B _____

C _____ D _____

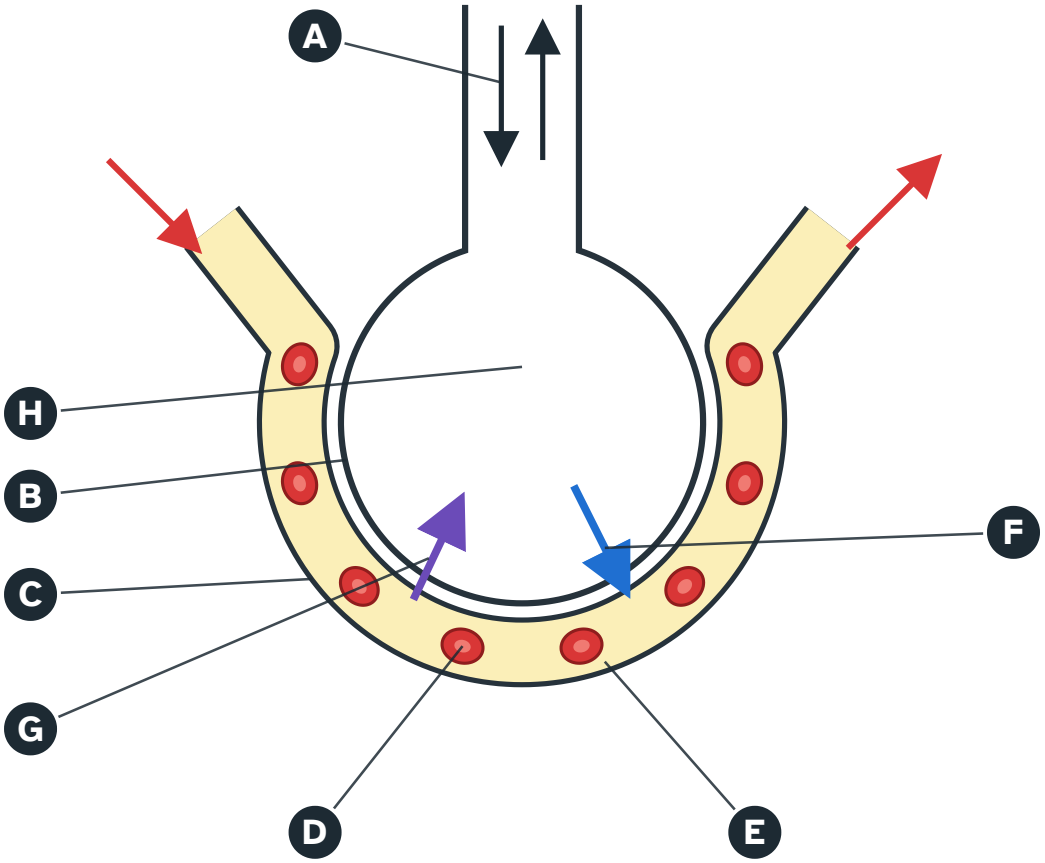
E _____ F _____

G _____ H _____

I _____ J _____

K _____ L _____

1.2 Test yourself: label the air sac



A _____ **B** _____

C _____ **D** _____

E _____ **F** _____

G _____ **H** _____

ANSWERS (COVER THESE UP)

A entrance to nose
B entrance to mouth
C larynx (voicebox)
D rings of cartilage
E trachea (windpipe)
F bronchus
G bronchiole
H air sacs (alveoli)
I lung
J rib
K intercostal muscles
L diaphragm

A air moving in
B thin wall of air sac
C thin wall of capillary
D red blood cell
E blood plasma
F diffusion of oxygen
G diffusion of carbon dioxide
H air inside sac

Top grid: respiratory system. Bottom grid: air sac.